



FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 01-08-2026

In addition to Part I (General Handout for all courses appended to the timetable), this portion gives further details regarding the course.

Course Number : **PHA F315**

Course Title : **PHARMACEUTICAL FORMULATIONS II**

Instructor-in-charge : **Ankit Jain (ankit.j@pilani.bits-pilani.ac.in, 3170-M)**

Lab Instructors : **Sweta Acharya, Yash Patidar, Yashika Tomar, Vagesh Verma, Shivangi Nema, Kumbhar Pragati Ramesh, Desai Vaibhavi Meghraj**

1. Course description: Physical, chemical, and biopharmaceutical considerations in formulations, absorption, distribution, and elimination of drugs; pharmaceutical additives; formulation, manufacturing, packaging, quality control, and stability of solid dosage forms, semi-solid dosage forms, liquids dosage forms, and sterile preparations including parenteral suspensions, emulsions, depot formulations; microencapsulation and controlled release formulations; targeted drug delivery systems; multi-unit particulate systems; transdermal, gastroretentive, and nasopulmonary drug delivery systems; aerosol products and packaging.

2. Scope & Objective of the Course: It is a course intended to give students insight into the problems involved in making drug dosage forms. It brings together the various sciences involved in developing pharmaceutical product formulations and evaluating their quality. The laboratory component involves some fundamental exercises in the evolution of products such as tablets, ointments, solutions, parenteral products, sustained-action medications, etc. It also imparts knowledge of newer dosage forms.

3. Course learning outcomes:

CLO1. Explain the principles of preformulation, formulation, manufacturing, packaging, quality control, and stability of conventional pharmaceutical dosage forms, including solid, liquid, semi-solid, and sterile preparations.

CLO2. Apply pharmaceutical formulation principles and standard manufacturing techniques to prepare and evaluate conventional and modified-release dosage forms, while adhering to Good Manufacturing Practices (GMP) and quality requirements.

CLO3. Compare and evaluate the design, advantages, limitations, and pharmaceutical applications of novel drug delivery systems, including controlled-release, targeted, transdermal, gastroretentive, multiparticulate, nasopulmonary, and aerosol dosage forms.

4. Textbooks and references:

TB1. Ansel's Introduction to Pharmaceutical Dosage Forms.

Reference Books:

RB1. Lachman, Liebermann & Kanig: Theory and Practice of Industrial Pharmacy, 3rd Edition.

RB2. Remington's Pharmaceutical Science

RB3. Swarbrick, Current Concepts in Pharmaceutical Sciences & Biopharmaceutics.

5. Course Format & Weekly Workflow

(a) Course Plan:

Wk no.	Module No.	Description	Reference	Learning Outcome
I	1	Formulation – an introduction	TB. Ch. 1	Basic principles of formulation
I	2	The drug delivery systems & classification of dosage forms	TB. Ch. 2	Understanding different dosage forms and their characteristics
I	3	Components of a dosage	TB. Ch. 2	Understanding formulation ingredients and



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		form		their effect
II	4	Design of dosage forms	RB2 (i)	How a dosage form is formulated and how to modify it for better action
II	5	Physical properties of drugs & formulation	TB. Ch. 3	Understanding how the physical characteristics of a drug and formulation component affect the formulation
II	6	Chemical properties of drugs & formulation	TB. Ch. 4	Understanding how the chemical characteristics of a drug and formulation component affect the formulation
III	7-8	Biopharmaceutical characteristics of drugs	TB. Ch. 5	Understanding how the physicochemical characteristics of a drug and formulation component affect bioactivity
III-V	9-13	Tablets and pellets	TB. Ch. 13 RB 1	Understanding of formulation aspects, manufacture of tablets and pellets
V-VI	14-16	Capsules	TB. Ch. 14	Formulation design and manufacture of capsules
VI-VII	17-19	Parenteral products	TB. Ch. 21	Types of Parenteral and their Formulation Aspects: Understanding the design and requirements of manufacturing facilities for sterile dosage forms
VII	20	Ocular dosage forms	TB. Ch. 22	Formulation and quality control of ophthalmic products.
VII-VIII	21-23	Sustained-release dosage forms	TB. Ch. 23	How can drug release from formulation be slow and controllable?
VIII-IX	24-26	Coating & Microencapsulation	TB. Ch. 32	Coating process and development of microencapsulated dosage form.
IX	27	Transdermal drug delivery systems	TB. Ch. 24	How to formulate a transdermal drug delivery system
IX-X	28-30	Targeted Drug Delivery Systems	TB. Ch. 25	Designing targeted drug delivery systems
XI	31-32	Other novel drug delivery systems	TB. Ch. 26 & 29	Designing various novel drug delivery systems
XI	33	Quality control of dosage forms	TB. Ch. 32	Quality control of dosage forms
XII	34	CGMP and Process Validation	RB3	CGMP and Process Validation

(b) Practicals:

Name of Experiments	No. of Practical Hours	Refer
Formulation of different dosage forms: Tablets, liquids, semisolids, and injectables	13 Lab Hours (each of 3 hours)	Different chapters of textbook

List of Experiments

Dosage Forms	Drugs (with dosage forms)	Quality Control Tests
Solid Oral (6 labs)	Paracetamol (Tablets)	QC tests for granules: Bulk density, Tapped density, Flow properties of granules
	Aspirin (Dispersible)	



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	Tablets)	QC tests for tablets: Physical observation of tablets (color, shape, capping, lamination), Hardness, Weight variation, Friability, Disintegration time, and Dissolution testing (demonstration).
Liquid orals (3 labs)	Propranolol (Solution)	Physical observation (color, clarity, suspended particles), Weight per ml, sedimentation rate, redispersibility (if applicable), Flowability, pH, and viscosity.
	Metronidazole (Suspension)	
	Amoxicillin (Reconstituted powder-based suspension)	
Semi-solids (2 labs)	Tobramycin (Ointment)	Physical observation (color), grittiness, ease of extrusion from the tube, spreading efficiency, ease of removal, and texture.
	Diclofenac sodium (Gel)	
Injectable (1 lab)	Calcium gluconate	Clarity, Particulate matter, Color, Leakage, Seal, Weight per ml, Sterility, Pyrogen test.
1 Lab	Demonstration	Fluidized bed processor, Spray dryer, Dissolution apparatus with autosampler

Writing Instructions for Record

Right Side of the Record:

- **Aim**
- **Materials and Apparatus**
- **Preformulation studies**
 - Physiochemical properties (solubility, pK_a , etc.)
 - Stability
 - Storage
 - Dose
 - Therapeutic use
- **Theory**
- **Procedure:** in brief, for both the formulae (stepwise)
- **Results and Observations**
- **Discussions:** Compare both the formulations
- **Conclusions**
- **References**

Left Side of the Record:

Formulae for both formulations (in tabular form as given below):

S.N.	Name of ingredient	Quantity per tablet or 5 ml in case of liquid, or % w/v in case of injectables, or % w/w in case of semisolids	Quantity per batch	Rational
1.	-----	-----	-----	-----
2.	-----	-----	-----	-----

- Write the ingredient's names as per their order of addition to the dosage form.
- **Batch size:** Tablet: 50 tablets, Liquid Oral: 50 ml, Semisolids: 25 gm, Injectable: 2-3 ml
- **Calculation and Observations (if any)**



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Contents to be written on the label

- **Name of product with specification:** I.P./ B.P./ U.S.P.
- **Category/ Therapeutic class:**
- **Quantity:** quantity per unit, or 5 ml in case of liquid, or % w/v in case of injectables or % w/w in case of semisolids
- **Batch No. / Lot No.:**
- **Storage conditions:**
- **Dose:**
- **Mfg. date:**
- **Expiry date:**
- **License number:**
- **Cautions/ Warnings (wherever applicable):**
- **Name and address of manufacturer:**

General Instructions for Lab

- ✓ **NO** makeup will be granted for the lab component[§].
- ✓ Students should attend every lab with a completed record and a rough record (dedicated to the lab of this course).
- ✓ Preformulation studies should be carried out well in advance, and two tentative formulae should be prepared based on the preformulation study.
- ✓ Students coming for laboratory sessions without proper preformulation data would be considered absent.
- ✓ Instructors will approve the formulae.
- ✓ Label (with all details) should be prepared for any one product.

[§]Lab components will consist of day-to-day work & viva (30%), Mid-Semester Exam (25%), and Comprehensive Exam (45%).

6. Evaluation Scheme:

Evaluation Component	Weightage (%)	Duration	Date & Time	Nature of components
Continuous assessments (Quizzes, assignments, etc.)	15	TBA	TBA	CB
Laboratory Components	30	Continuous	Lab hours	OB+CB
Mid-Sem Exam	20	90 min.	As announced in the Timetable	OB
Comprehensive Exam	35	180 min.	As announced in the Timetable	CB

Grading would be done using the bunching procedure. In borderline cases, subjective judgment will be used to award the grades. Awarding all eight grades (i.e., from A to E) is not mandatory. Subjective judgment based on attendance at lectures and tutorials, participation in quizzes, students' involvement in the course, and performance in the lab component would be used to award grades. The student shall not be considered "exposed" to the course unless he/she demonstrates appreciable skill in both the laboratory and theory components of the course and through classroom participation. Attending tutorials and appearing in surprise quizzes are very important. Students who have not participated in quizzes, submitted assignments, or completed any tests/labs/comprehensive exams will not be considered as having completed/exposed to the



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course. Students are also expected to attend classes regularly to gain proper exposure and to evaluate their knowledge.

Criterion for NC: 30% of the average of the top three performers or 40% of the median, whichever is lower, or a suitable modification at the discretion of the IC with the concurrence of the Department.

7. Attendance Policy: Each student is expected to attend all classes and participate in all evaluation components.

8. Chamber Consultation Hours: To be announced in class.

9. Make-up policy: Makeup will be given only for genuine cases (accompanied by proof) for the Mid-semester or comprehensive exam only.

10. Notices: All notices concerning the course will be emailed to the students.

Instructor-in-charge
PHA F315